

Article D4: Mobility Innovation Series (MIS)

AutoDrive Challenge II introduces the Mobility Innovation Series (MIS) to participating teams. This series of challenges will require teams to develop innovative, multi-disciplinary solutions to real-world problems. The goal of these challenges is to introduce the many facets of autonomous & electrified vehicle development, as well as supplement each team's individual vehicle development cycles. These challenges may facilitate inclusion of many non-engineering focused students, such as those in economics, marketing, social media, and management, and still require engineering input.

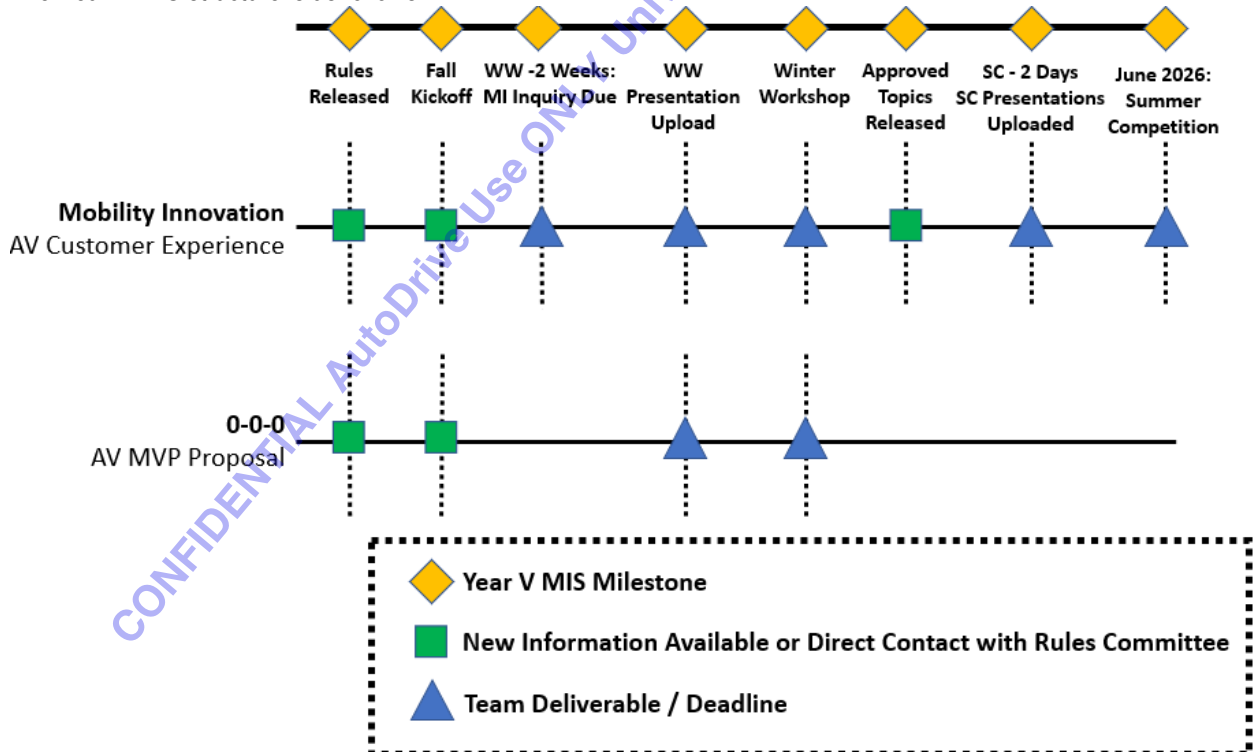
MIS.1.1 Mobility Innovation and 0-0-0 Challenges

Year V will have one Mobility Innovation challenge and one Zero-Zero-Zero (0-0-0) challenge. Innovation challenges will focus on a variety of aspects, ranging from vehicle-to-vehicle interaction, vehicles to the environment, data collection, cybersecurity, and others. 0-0-0 challenges, on the other hand, will align with on General Motor's mission of Zero Crashes, Zero Emissions, and Zero Congestion by 2030.

MIS.1.2 Structure and Timing

Year V's Mobility Innovation Series will follow a similar structure to year IV. Teams are expected to participate in each challenge in addition to other deliverables as outlined in the AutoDrive Challenge II Rules.

The Year V MIS structure is as follows:



Timing overviews will be provided within each Challenge's ruleset.

MIS.1.3 Deliverables and Scoring

Each Mobility Innovation challenge will have several deliverables which vary per challenge. Year V MIS Challenges will total 75 points: 45 points for the Innovation Challenge and 30 for the 0-0-0 Challenge. See each individual challenge's overview for scoring details.

MIS.1.4 Violation of Rules

In addition to the rules outlined for these challenges, the rules of ethics and academic integrity apply, meaning that plagiarism, using faculty in the writing of papers or presentations, and theft of other team's ideas will be considered a violation and the team eliminated from the competition. All sources for information, data, research, and reports must be cited using MLA or AMA style of writing, format and citations. Any violation of these rules either intentionally or unintentionally will result in an appropriate sanction that includes assessment of a penalty, disqualification from the competition and/or the return of prize money. The imposition of sanctions and penalties is within the sole discretion of the Mobility Innovation Rules Committee ("MIRC") and is not subject to discussion, debate, or challenge.

Article D5: Mobility Innovation Event (MIE) Year V Innovation and 0-0-0

MIE 1 Innovation: AV Customer Experience

MIE.1.1 Problem Statement

OEMs are always looking for opportunities to increase their market share, whether through the introduction of attractive new vehicles or the incorporation of advanced features. As autonomous vehicles evolve, OEMs need to ensure that customers are not only open to AVs but also enthusiastic about using them, particularly since some may be reluctant to transition from traditional driver-operated vehicles. A key strategy for promoting this adoption is to improve the customer experience and set themselves apart from competitors.

MIE.1.2 Challenge & Considerations

Teams will take the position of a start-up focused on improving the AV experience for customers. Your team is challenged to present a product pitch at Summer Competition to industry professionals as if you intend to gather seed funding. To consider investing in your product, potential investors need to understand your team's motivation, the details of the product, and your business plan. Your presentation should open with an engaging elevator pitch and close with a strong appeal for support. Teams may target Ride Share, Personal, or both types of AVs.

Notes:

- The proposed product/feature must be present on the AV, either physically or through the AV's internal software. For example, reward-based app/systems designed to encourage AV use will not be permitted.
- Teams must propose a novel product to improve customer experience. In other words, the solutions should not already be commonly utilized or self-evident within the AV space.

MIE. 1.3 Challenge Definitions

- OEM: Original Equipment Manufacturer. The final assembler of the complete vehicle, whether it is a passenger, commercial, or off-road vehicle.
- AV Customer Experience: Encompasses the overall journey and interactions a customer has with an AV from vehicle entrance to exit. Customer experience can be characterized by many factors including ease of use, satisfaction, and emotional connection with the brand.
- KPI: Key Performance Indicator. A measurable value that demonstrates how an organization achieves self-determined key objectives.

MIE 1.4 Deliverables

Teams shall be presenting their solution proposal to a live audience at the Summer Competition. Teams will be scheduled in blocks of 30 minutes. Teams are permitted to present for a maximum of 15 minutes, with an additional 10 for Q&A from the judges and audience.

Presentation content shall include and address, at minimum, the following:

- Engage the audience: Start with a 60 second elevator pitch that captures the judge's attention.
 - Series resources and reference guides will be provided at Fall Kickoff.
- Define the problem statement: Identify and justify a product or feature that improves the customer's AV experience. This can solve an existing problem or enhance one's experience.
 - Justification should include background research and measurable data that establishes a problem to be solved.
- Pitch the solution: Sell the audience on your idea. Pitches should address the following:
 - Stakeholders & Impacts
 - Who are the primary beneficiaries?
 - How does the product or feature help enhance their experience?
 - Benchmarking & Market Assessment
 - Has anyone else tried to address the problem?
 - If not, why hasn't the problem been addressed yet? What challenges currently exist?
 - Team's Solution: Technical & Customer-facing Details
 - What is the team's solution?
 - What differentiates your product from others?
 - Share a system diagram and/or CAD/simulation that explains how the product works.
 - How does it integrate into the vehicle?
 - Explain how the customer would experience the product / feature.
 - Business Plan
 - What is your business plan?
 - How does this make money, draw more customers, or meet other self-defined KPIs?
 - How much does the solution cost to implement? This should include estimates for hardware, software, and engineering design & development.
- Leave a strong impression:
 - Concluding statements should outline final thoughts and end with a strong appeal for funding and support.

MIE 1.4 Timing and Enablers

The Mobility Innovation Rules Committee has set forth the following as a high-level project management plan. Note that there are deliverables that all teams must comply with.

Activity	Date or Time Period	Recommended Actions and Deliverables
Rules Provided and Research	Fall 2025	Recommended Action(s): Review the Rules prior to Fall Kickoff.

Fall Kickoff	Fall Kickoff	Mobility Innovation Rules Committee will answer questions and provide additional guidance to teams. Recommended Action(s): Attend Fall Kickoff with questions prepared.
Idea Generation and Design	Fall Kickoff - Winter Workshop	Deliverable(s): Submit a Rules Inquiry titled "Mobility Innovation Proposal: <Team Name>". The Mobility Innovation Rules Committee will respond with a topic approval or additional questions within 10 business days. If you do not receive a response, please have your mentor reach out to MIRC. Upload a 1 page presentation summarizing each topic proposal via the Autodrive portal. If you are submitting more than 1 topic for consideration, each topic should have a separate page (i.e., a PDF with 3 pages should have 3 separate topics at 1 page each). Note: topic summaries exceeding 1 page and/or with modified formatting will be automatically denied. They can be resubmitted in the prescribed 1 page summary. Please see section "Inquiry Template" for a suggested inquiry format.
Topic Inquiries	Winter Workshop - 2 Weeks	Deliverable(s): All teams shall have their topic inquiries & topic summaries submitted.
Winter Workshop Presentation Upload	Winter Workshop - 2 Days	Deliverable(s): All teams shall upload their intended presentation 2 days prior to the start of Winter Workshop.
Winter Workshop	February 2026	Deliverable(s): Each team shall present a non-graded, 15-minute overview of their status in preparation for Summer Competition to the AutoDrive Rules Committee. This presentation replaces both the Mentor progress updates and Mentor evaluations previously held in Year III and IV. The intent of this review is for teams to demonstrate that they are prepared for Summer Competition. Teams' design solutions should be close to completion at this point.

Approved Topics Release	Winter Workshop + 1 Week	Mobility Innovation Rules Committee will release a generalized list of approved topics to Teams. Any teams still pending an approved topic can review this list to be informed of what is already approved.
Presentation Refinement	February 2026 – June 2026	Recommended Action(s): Teams presenting at Summer Competition should focus on finalizing, practicing, and honing their presentations. Teams are recommended to review and solicit feedback on their presentation with various audiences.
Presentation Upload	2 days prior to Summer Competition Start	Deliverable(s): Presentations will be completed and uploaded to the AutoDrive portal. No edits will be permitted after the submission period.
Summer Competition	June 2026	Deliverable(s): Presentations will be given live. The expectation is that a limited number of team members will present. Presentation order and times will be communicated at least 1 day prior.

MIE 1.5 Inquiry Template

Below is an image of the Inquiry Template. An editable version will be made available in Series Resources.

Team Name:
Date:

Replace With Title of Proposed Topic

Please use this space to summarize your proposed topic. Keep the summary concise. No more than one page should be used per proposal. A team may propose more than one topic/idea.

NOTE: Formatting of this document may not be changed (spacing, font size, etc.).

Example:

- 1-2 sentences contextualizing your proposed topic.
- 1-3 sentence explanation of proposed topic.

MIE 1.6 Scoring

To ensure consistency and fairness in judging at Summer Competition, a set panel of judges will score each team's presentations. Scoring of these deliverables will total 45 points.

Category	Maximum # Points / 45	Notes
Elevator Pitch	5	Captures the attention of the judges/audience, and clearly highlights the position of the team. It should include an introduction, explanation of company/product, a unique value proposition, and call to action.
Problem Statement	5	The proposal identifies and justifies a product or feature that improves the customer's AV experience.
Benefits to Stakeholders	10	The presentation should highlight the benchmarking performed by the team, provide a compelling reason for judges to invest, and highlight the benefit to the customer.
Technical Complexity	10	The selected proposal should include a technical explanation and differentiate itself from existing solutions.
Feasibility of Product/Feature	10	The proposal should be practical, financially viable, and contain a business plan.
Professionalism and Persuasion	5	The presentation is perceived as professional, the team appears as prepared to present, and judges are persuaded to invest in the proposal.

MIE 1.7 What Happens If ...?

In an effort to address questions and align rules interpretations across Teams, the following hypotheticals have been addressed in a "What happens if ...?" format:

What happens if...	Response
My team has started investigating a topic prior to MIRC approval?	MIRC reserves the right to change the team's direction. Get your topic approved early.
My team doesn't have an approved topic by the Winter Workshop - 2 weeks deadline?	Teams in this situation will receive a flat 5% penalty. Note: Teams will still need to obtain an approved topic via inquiry from MIRC if they do not have one by Winter Workshop.
My team doesn't have material ready to present at Winter Workshop?	We'll discuss presentation techniques during the designated timeslot. Teams in this situation will receive a flat 5% penalty.
My team doesn't have an approved topic before Summer Competition?	Teams will incur a 5% flat point penalty per week -- up to 20% total -- leading up to the presentation upload deadline. Penalties will be assessed by when teams submit their topic inquiry & upload their topic summary that results in an approved topic. Note: This creates a potential "grace period" for teams to submit their topic inquiry after Winter Workshop without accruing the additional 5% penalty per week.

My team wants to make changes to our Summer Competition presentation after the submission portal has closed?	No changes will be permitted. The presentation upload date has been adjusted closer to Summer Competition to address this concern. No exceptions will be permitted.
My team uploads the Summer Competition presentation late but before the first Summer Competition presentation?	A flat 20% point penalty will be assigned.
My team uploads the Summer Competition presentation after the first Summer Competition presentation?	If the presentation is not uploaded by the beginning of the first presentation at Summer Competition, the team may still choose to present but will receive an automatic 50% flat-point reduction.
My team is penalized, and I want to appeal?	Refer to MIS 1.4 Violation of the Rules. MIRC will elaborate on penalties during the final point submission.
Penalties results in a negative score?	Teams in this situation will receive a score of zero.

MIE 1.8 Penalties

The following chart summarizes penalties for this challenge. Teams that meet the prescribed deadlines and follow the rules will not incur penalties.

Flat Penalty Percentage	Flat Point Amount	Deadline	Criteria
5	2.25	Winter Workshop – 2 weeks	Topic Inquiry Due
5	2.25	Winter Workshop – 2 days	Pre-review presentation not uploaded before Winter Workshop
5	2.25	Summer Competition Upload – 4 weeks	Late Topic Inquiry Due
5	2.25	Summer Competition Upload – 3 weeks	Late Topic Inquiry Due
5	2.25	Summer Competition Upload – 2 weeks	Late Topic Inquiry Due
5	2.25	Summer Competition Upload – 1 week	Late Topic Inquiry Due
20	9	Summer Competition Upload	Presentation uploaded after deadline but before Summer Competition presentations begin
50	22.5	Summer Competition First Presentation	Presentation uploaded after first team presents

Note: MIRC will accelerate topic inquiry responses after Winter Workshop to a 3-business day turnaround. Penalties will be accrued on a week-by-week basis.

MIE 2 0-0-0: AV Minimum Viable Product (MVP) Proposal

MIE 2.1 Problem Statement

A Minimum Viable Product (MVP) is a product that contains the bare minimum content and features that produce the desired outcome for its customers. The goal of an MVP is to make considerations for resources such as time, complexity, cost, ease of implementation, etc. This approach helps in reducing risks and ensuring that the final product aligns with market demands. Any MVP must still comply with applicable regulations to be used on public roads.